

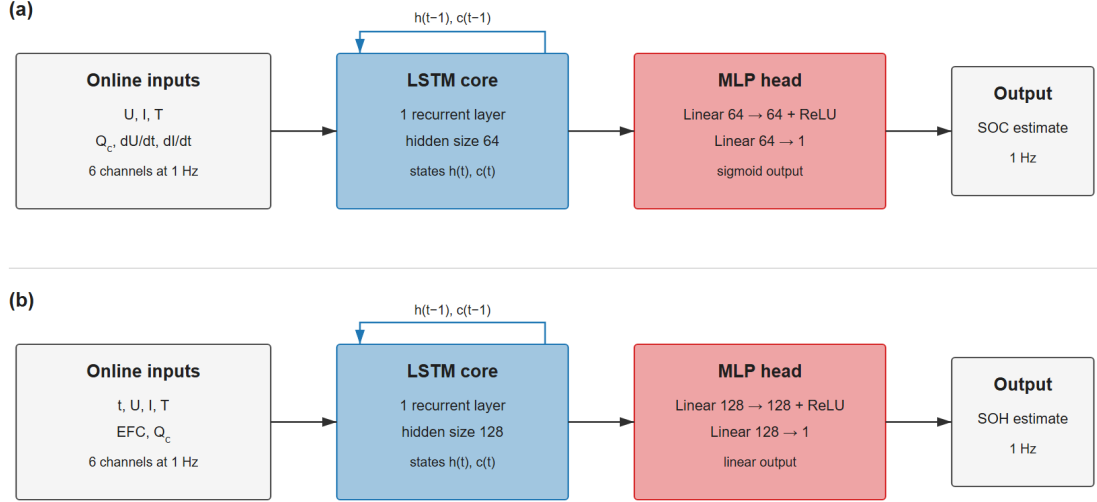


Figure 7.2: Task-specific architectures of the two embedded estimators. (a) The SOC estimator combines six online input channels with a single-layer LSTM of hidden size 64 and an MLP head with sigmoid output. (b) The SOH estimator processes its six online input channels with a single-layer LSTM of hidden size 128 and an MLP head with linear output. Both models operate statefully on the sample-wise 1 Hz stream and propagate their hidden and cell states between consecutive updates.

For input dimension $D = 6$, hidden size H , and MLP width M , the weight-related multiply-accumulate count of one streaming update is

$$N_{\text{MAC}}(H) = 4H(D + H) + HM + M. \quad (7.9)$$

This count excludes bias additions, activation functions, indexing, and memory traffic. If a fraction p of the recurrent channels is removed, the retained dimension is $H_p = (1 - p)H$. The corresponding analytical reduction follows as

$$\begin{aligned} R_{\text{MAC}}(H, p) &= 1 - \frac{N_{\text{MAC}}((1 - p)H)}{N_{\text{MAC}}(H)} \\ &= \frac{4(2p - p^2)H^2 + p(4D + M)H}{4H^2 + (4D + M)H + M}, \\ \lim_{H \rightarrow \infty} R_{\text{MAC}}(H, p) &= 2p - p^2. \end{aligned} \quad (7.10)$$

The quadratic recurrent contribution is reduced more strongly than the linear input